

AI-Assisted Criminal Investigations: Enhancing Testimony Analysis and Case Correlation

Incident Correlation and Pattern Identification

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Final Report

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Declaration

I declare that this is my own work, and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text. Also, I hereby grant to Sri Lanka Institute of Information Technology the non-exclusive right to reproduce and distribute my dissertation in whole or part in print, electronic or other medium. I retain the right to use this content in whole or part in future works (such as article or books)

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The above candidate has carried out research for the bachelor's degree Dissertation under my supervision.

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Abstract

This research addresses critical limitations in traditional criminal investigations by developing an AI-assisted system that enhances testimony analysis and dynamic interrogation processes. Conventional interrogation methods rely extensively on investigator expertise, which can be affected by cognitive biases, fatigue, and subjective judgment, leading to inefficiencies and compromised accuracy. Leveraging advances in Natural Language Processing (NLP) and transformer-based Large Language Models (LLMs), this system introduces dynamic question generation tailored in real-time to suspect responses. It integrates real-time analysis of testimony to detect inconsistencies, contradictions, and missing information, thereby providing investigators with actionable and contextually relevant follow-up questions. The system further incorporates multimodal behavioral analytics including emotion and stress indicators from speech and facial cues to enrich interrogation insights. Architected as a scalable full-stack application with React/Next.js frontend, MongoDB backend, and APIs interfacing with LLMs, the platform supports secure, multi-session interrogations with an intuitive investigator interface. Comprehensive feasibility studies, field visits, and data collection from Sri Lankan law enforcement ensured alignment with operational, legal, and ethical requirements. Experimental evaluation demonstrated the LLM's superior semantic relevance for question generation against expert-curated benchmarks. The system maintains ethical safeguards such as human oversight, data anonymization, and bias monitoring to ensure fairness and judicial compliance. This AI-assisted interrogation assistant promises to enhance investigation efficiency, accuracy, and ethical standards while accelerating case resolution and strengthening the criminal justice process.

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List of Abbreviations

AI - Artificial Intelligence

Api - Application programming Interface

BERT - Bidirectional Encoder Representation From Transformers

CRUD - Create , Read, Update ,Delete

DB - Database

Json - JavaScript object Notation

ML – Machine learning

NLP - Natural language processing

UI – User Interface

UX – User Experience

1. Introduction

The primary objective of this research is to enhance the efficiency and reliability of criminal investigations by developing an AI-driven system that automates testimony analysis, incident correlation, and victim risk assessment. Traditional investigative processes in Sri Lanka and worldwide rely heavily on manual review of police reports, case files, and witness testimonies. This approach is often time-consuming, prone to human error, and constrained by the cognitive limits of investigators, leading to overlooked connections and inconsistencies. The proposed system seeks to address these shortcomings by integrating machine learning, natural language processing, and heuristic scoring techniques into a unified decision-support framework.

At the core of the system is a Random Forest classifier trained on structured case data, including victim and perpetrator demographics, type of violence, and evidence indicators. This model predicts the likelihood of criminal responsibility, offering investigators a quantifiable measure of offender risk. In parallel, an NLP module leverages transformer-based embeddings to analyze witness testimonies, enabling semantic similarity comparisons across new and historical cases. By moving beyond keyword searches, the system can detect subtle narrative overlaps or contradictions, improving the detection of repeated or related incidents. Complementing this, a lightweight keyword correlation engine provides lexical matching for rapid identification of surface-level case similarities.

A distinctive feature of the system is the victim risk assessment module, which applies rule-based scoring to evaluate vulnerability factors such as age, gender, recurrence of violence, injuries, and relationship to the perpetrator. This ensures that the investigative process is balanced, considering not only offender likelihood but also the safety and protection needs of victims. To ensure interpretability, an explainability component highlights the key factors influencing each prediction, while verdict logic consolidates the outputs into intuitive classifications: Likely Criminal, likely Victim, or Inconclusive.

The system is implemented using a modern technology stack comprising MongoDB for case storage, FastAPI for backend orchestration, and a Next.js dashboard for visualization. Investigators can input case identifiers and receive real-time outputs including predictions, risk scores, verdicts, explanations, and links to similar historical cases. This integration of structured and unstructured data analysis into a single pipeline represents a significant step towards modernizing investigative workflows.

Overall, the proposed system aims to improve the speed, accuracy, and transparency of criminal investigations. By bridging the gap between manual investigative practices and data-driven analytics, it provides law enforcement with a scalable and explainable tool for decision support. In doing so, it contributes not only to improving case resolution but also to advancing the responsible adoption of artificial intelligence in the sensitive domain of criminal justice.

1.1 Background Literature

Crime investigation has historically relied on a combination of documentary evidence, testimonies, and physical records. Investigators are tasked with piecing together large volumes of fragmented information, ranging from witness statements and police notes to forensic reports. This process is often labor-intensive and prone to oversight, especially when datasets grow beyond the capacity of manual cross-referencing. As societies expand and digital evidence becomes more prevalent, law enforcement agencies face increasing difficulty in managing the speed, scale, and complexity of modern investigations [1].

The introduction of computational methods into policing has created new opportunities to improve the accuracy and timeliness of investigations. Early systems emphasized digital storage and retrieval of police records, but these provided little support for deeper pattern analysis. More recent developments in artificial intelligence (AI) and machine learning (ML) offer techniques capable of extracting hidden connections, identifying anomalies, and providing real-time decision support [2]. By applying algorithms to testimony, case histories, and contextual data, it is possible to highlight correlations that may otherwise remain unnoticed.

Behavioral analysis has also become an important strand of investigative science. The interpretation of verbal and non-verbal cues, historically conducted by trained investigators, can now be supported by computational methods. Advances in natural language processing (NLP) allow systems to examine witness testimonies at scale, uncovering contradictions, similarities, or suspicious patterns [3], [4]. Combined with emotional and psychological modeling, these technologies help bridge the gap between raw testimony and actionable insights.

Furthermore, law enforcement increasingly requires tools that go beyond static reporting. Investigative processes must accommodate real-time alerts, enabling officers to react promptly to anomalies or emerging threats. For example, an incident involving repeated threats from a suspect may be flagged earlier if previous cases exhibit similar behavioral markers [5]. Such early detection is particularly valuable in preventing escalation, protecting victims, and allocating limited resources effectively.

The current project builds upon these developments by proposing a system that integrates machine learning classification, semantic testimony analysis, incident correlation, and victim risk assessment into a single investigative support tool. By combining structured data (ages, genders, violence categories) with unstructured data (witness summaries, incident narratives), the system aims to reduce human error, accelerate case resolution, and modernize investigative workflows.

1.2 Literature Review

The academic literature on AI-driven criminal investigations has grown significantly in recent years, reflecting the increasing demand for technologies that can handle complex case data, testimonies, and behavioral evidence. Researchers have explored multiple domains, including case correlation, natural language processing for testimony analysis, victim risk assessment, and real-time decision-making in policing. Collectively, these studies confirm the potential of AI to reduce investigative workload and improve accuracy, but they also highlight critical limitations that motivate the present research.

One of the most widely studied areas is the use of machine learning for incident correlation and case linking. Traditional investigative practices rely on manual cross-checking of reports, which is inefficient when databases become large and fragmented. Johnson et al. [6] demonstrated that automated forensic tools can effectively cross-match police records and testimonies across multiple cases, detecting relationships that human investigators might miss. Similarly, Lee et al. [7] applied clustering and neural network approaches to map relationships between crimes based on offender profiles and geographical information. Their work confirmed that algorithmic methods can uncover hidden trends and link seemingly unrelated incidents, thereby enabling faster resolution of repeat offenses and serial cases. However, many of these tools focus on structured attributes, such as location or offender type, and neglect unstructured narratives like testimonies and incident summaries.

Natural language processing (NLP) has emerged as a promising approach to address the challenge of analyzing unstructured testimonies. Transformer-based models such as BERT have set new benchmarks for semantic text representation [3]. By encoding narratives into dense vector embeddings, these models allow systems to compare testimonies on the basis of meaning rather than simple keyword overlap. Smith et al. [8] explored this potential by applying NLP to police testimonies, showing that transformer-based models could detect contradictions and similarities that were not captured by traditional keyword search techniques. Other deep learning approaches, such as recurrent convolutional neural networks [4] and very deep convolutional models [7], have also demonstrated effectiveness in text classification tasks relevant to law enforcement. These findings collectively show that semantic similarity scoring can support investigators in assessing credibility and prioritizing leads. Nonetheless, most prior studies remain confined to offline or experimental settings, without demonstrating real-time integration into investigative workflows.

Another strand of research emphasizes the importance of real-time AI in policing. Brown et al. [9] argued that static reporting systems are inadequate for modern security needs, and highlighted the value of systems capable of processing live data streams. Their research on AI-driven crime monitoring systems combined CCTV feeds, incident reports, and dispatch records to deliver proactive alerts for law enforcement. Such dynamic processing enables agencies to allocate manpower more effectively and respond to

emerging threats before they escalate. However, the majority of these systems focus on large-scale surveillance and crime monitoring, while few directly integrate with the day-to-day tasks of investigators reviewing testimonies and case reports.

The adaptability of AI systems has also been recognized as essential in addressing evolving crime trends. Zhang et al. [10] proposed adaptive AI models that continuously refine their predictions as new cases are introduced, ensuring that the system remains relevant in the face of changing offender behaviors. This is particularly important in contexts such as cybercrime or organized crime, where tactics evolve rapidly and static models quickly become outdated. Their work highlights the necessity of incorporating continuous learning mechanisms into investigative systems to ensure long-term accuracy. However, integrating adaptability into a system that must also remain interpretable and reliable for legal use remains a challenge.

Victim-centered analysis has received less attention compared to offender profiling, though some significant contributions exist. Campbell [12] identified demographic and situational risk factors that increase the likelihood of repeat victimization or femicide, such as the use of weapons, prior history of abuse, and repeated incidents. Lakshan et al. [13] developed a real-time deception detection system that incorporated multimodal cues, demonstrating the potential for behavioral analytics in supporting investigations. Baltrūnienė [14] further emphasized that most AI-driven tools remain offender-centric and argued for the development of systems that can balance offender risk prediction with victim vulnerability assessment. This gap underscores the need for tools that protect victims proactively, rather than focusing solely on identifying suspects.

Finally, explainability has been identified as one of the most pressing concerns in the adoption of AI in law enforcement. While models such as Random Forests and neural networks can produce highly accurate predictions, their opacity often prevents investigators and courts from fully trusting their outputs. Doshi-Velez and Kim [15] advocated for a rigorous framework for interpretable machine learning, while Miller [16] discussed how explanation design can be informed by insights from the social sciences. Rigano [17] highlighted the reluctance of law enforcement agencies to adopt AI systems without clear justification for their predictions, noting that explainability is a prerequisite for building trust. Yet, despite these calls, few existing systems provide contextual reasoning or user-friendly explanations alongside their predictions. Taken together, the literature confirms that AI methods can transform criminal investigations by enabling case correlation, testimony analysis, victim assessment, and real-time decision support. However, it also reveals important limitations: most systems focus narrowly on either structured or unstructured data, many operate only in experimental contexts without real-time integration, victim-centered modules remain underexplored, and explainability is often neglected. Addressing these gaps, the present research introduces a unified system that combines Random Forest prediction, BERT-based testimony similarity, keyword correlation, heuristic victim risk scoring, and explainability into a single investigative support tool.

1.3 Research Gap

Although recent advancements in artificial intelligence (AI) have introduced powerful tools for law enforcement, several limitations remain in current research that restrict their effectiveness in real-world criminal investigations. Existing literature has explored AI applications in domains such as predictive policing, forensic analysis, and deception detection [1]–[4]. However, most of these contributions remain siloed, focusing on individual tasks rather than delivering an integrated decision-support framework for investigators. A key limitation identified is the inadequate integration of structured and unstructured data. Many systems are restricted to either tabular features (e.g., age, gender, location) or textual narratives, but rarely combine the two into a unified analytical pipeline [5]. This gap limits the ability of AI systems to detect hidden connections that span across demographic details and semantic testimonies.

Another major challenge is the lack of real-time similarity detection in witness testimonies. While transformer-based NLP models such as BERT have been widely recognized for their semantic representation power [6], prior studies often evaluate them offline, focusing on classification tasks rather than deploying them interactively for testimony comparison. This leaves investigators without tools to dynamically detect contradictions or overlaps in narrative evidence. Furthermore, victim-centered risk assessment remains an underexplored area. Existing research overwhelmingly emphasizes offender profiling and predictive policing [7], [8], with minimal focus on quantifying victim vulnerability. As highlighted in criminological studies [9], demographic and situational factors such as repeated incidents, injuries, and domestic relationships play a crucial role in predicting victim safety. However, current AI systems rarely incorporate such scoring modules to inform preventive measures. The issue of explainability also poses a significant gap. Black-box models such as deep neural networks and ensemble classifiers provide accurate predictions but often fail to justify their outputs in investigator-friendly terms [10]. Without transparent reasoning or contextual references, law enforcement agencies remain reluctant to fully adopt AI systems in practice.

Finally, most research lacks an integrated platform that unifies prediction, similarity analysis, correlation, risk assessment, and explainability into a single interface. Studies tend to focus on isolated tasks, such as forensic analysis, deception detection, or predictive modeling, without considering end-to-end workflows. This fragmentation prevents investigators from efficiently leveraging AI across multiple dimensions of case analysis. To address these gaps, the proposed system integrates Random Forest prediction, BERT-based testimony similarity, keyword correlation, heuristic victim risk scoring, explainability, and web-based visualization into a modular and accessible platform. By combining structured and unstructured case data, delivering real-time analysis, and providing transparent outputs, the system bridges the disconnect between academic research and operational investigative needs.

Application Reference	Research A	Research B	Research C	Research D	Proposed System
Integration of Structured + Unstructured Data	✗	✗	✓	✗	✓
Real-Time Testimony Similarity (BERT-based)	✗	✓	✗	✗	✓
Automated Incident Correlation	✗	✓	✓	✗	✓
Victim-Centered Risk Assessment	✗	✗	✗	✗	✓
Explainability of Predictions	✗	✓	✗	✗	✓
Get Audio Records in Each Pages and Create Sinhala Audio Book	✗	✗	✗	✓	✓

Table 1 Research Gap

1.4 Research Problem

Modern criminal investigations generate vast amounts of both structured and unstructured data, including police reports, witness testimonies, forensic records, and historical case files. Traditionally, investigators rely on manual techniques to analyze this information, identify patterns, and connect related incidents. However, this process is slow, error-prone, and often results in critical details being overlooked. As the volume and complexity of case data continue to grow, law enforcement agencies face significant challenges in achieving timely and accurate resolutions.

A major difficulty lies in correlating incidents across different regions and timeframes to uncover concealed patterns of criminal behavior. Existing investigative tools are often fragmented, designed for narrow tasks such as biometric recognition, crime mapping, or forensic analysis. Few provide a consolidated solution capable of automatically analyzing case data to reveal meaningful links between incidents. Without such integration, investigators struggle to convert large datasets into actionable insights, leading to delays in case resolution and, in some cases, enabling offenders to evade justice.

Unstructured sources of information such as testimonies, interview transcripts, and case narratives remain particularly underutilized. Manually comparing these documents is not only time-consuming but also difficult due to variations in language and context. Current systems lack advanced natural language processing capabilities that can detect semantic similarities, contradictions, or evolving patterns in witness accounts. This limitation prevents investigators from making full use of critical qualitative evidence. Furthermore, investigators face difficulties in visualizing and tracking relationships between multiple cases. Existing systems provide limited or static views of case data, making it difficult to identify hidden connections or recurring behaviors across different incidents. The absence of real-time, AI-powered tools for correlation, similarity detection, risk assessment, and explainability represents a major research gap. Addressing this problem is essential to modernizing investigative workflows, improving the accuracy of case analysis, and ensuring the protection of victims and communities.

1.5 Research Objectives

1.5.1 Main Objective

The main objective of this research is to design and implement an AI-driven investigative support system that assists law enforcement agencies in analyzing criminal cases more effectively by integrating machine learning, natural language processing, incident correlation, victim risk assessment, and explainability into a unified platform.

The proposed system addresses the limitations of traditional investigative workflows, which rely heavily on manual review of police reports, testimonies, and forensic records. Such processes are often slow, inconsistent, and prone to human oversight, especially when datasets become large and unstructured. By combining structured attributes (such as victim and perpetrator age, gender, type of violence, and evidence presence) with unstructured narratives (witness testimonies, incident summaries), the system aims to automatically extract meaningful patterns, provide predictions, and highlight correlations that investigators may overlook.

Unlike existing fragmented tools that address isolated tasks, this system integrates multiple analytical modules into a single decision-support pipeline. It produces outputs such as criminal likelihood prediction, victim vulnerability scores, case similarity matches, and human-readable explanations, which are delivered in real time through a web-based dashboard. In doing so, the project seeks to modernize investigative workflows, reduce investigative delays, and improve both the efficiency and fairness of criminal justice processes.

1.5.2 Sub Objectives

- **Feature Extraction Module**

To develop a preprocessing pipeline that converts raw police reports into machine-readable features. This includes parsing victim and perpetrator demographics, encoding gender and violence types, detecting the presence of evidence, and calculating derived metrics such as age differences. The feature extraction module ensures consistent and standardized data for downstream machine learning models.

- **Criminal Prediction using Random Forest**

To design and train a Random Forest classifier that predicts the likelihood of criminal responsibility based on structured case data. The model generates a

probability score (expressed as a percentage) that indicates whether the perpetrator should be classified as likely criminal, thereby supporting investigators with quantitative decision-making.

- **Testimony Similarity using NLP (BERT Embeddings)**
To implement a semantic similarity module that uses transformer-based embeddings (BERT) to analyze unstructured narratives such as witness testimonies and incident summaries. This module computes cosine similarity between new and historical cases, highlighting related incidents even when different vocabulary is used.
- **Automated Incident Correlation**
To build a lightweight correlation engine that applies keyword matching and lexical overlap techniques to identify related cases. This complements semantic similarity by linking cases with shared keywords (e.g., names, locations, weapons), thus improving the breadth of correlation analysis.
- **Victim Risk Assessment Module**
To design a heuristic scoring mechanism that evaluates victim vulnerability. The module considers factors such as age, gender, injuries, recurrence of violence, and perpetrator relationship, producing a risk score categorized into low , Moderate , High , or Critical . This ensures that investigative focus includes victim safety alongside offender profiling.
- **Verdict Logic and Explainability**
To establish a transparent rule-based decision layer that integrates the outputs of prediction and risk scoring into intuitive verdicts (likely criminal, Likely Victim , or Inconclusive). Additionally, to implement explainability features that highlight key contributing factors and present top similar cases, providing investigators with human-readable justifications for system recommendations.
- **System Integration and Dashboard**
To integrate all modules into a FastAPI backend and develop a Next.js web-based dashboard that visualizes results in real time. The dashboard displays likelihood scores, victim risk assessments, correlated cases, and explanations, ensuring usability and accessibility for investigators.

2. Methodology

2.1. Requirements Gathering

The first stage of the research involved a careful requirement gathering process aimed at understanding the limitations of current investigative practices. Traditional investigation workflows depend heavily on human interpretation of witness statements, police notes, and behavioral records. Investigators often face large volumes of unstructured information that must be reviewed manually, making it difficult to identify contradictions or hidden connections between cases. This manual process is not only time consuming but also introduces the risk of human error and subjective bias.

In order to define the requirements for the system, existing investigative methods were reviewed along with real-world case workflows. The analysis revealed several recurring pain points. Investigators often struggle to cross-check witness statements, especially when different vocabulary is used to describe similar incidents. For example, one report may describe “domestic assault” while another describes “spousal violence,” both referring to similar acts but remaining unlinked in traditional databases. Another key issue is the lack of tools for assessing victim vulnerability. While much attention is given to identifying suspects, little structured support is provided for evaluating whether a victim is at high risk of repeated harm. Furthermore, investigators often receive little or no support in interpreting predictions made by machine learning models. Many existing systems provide numerical outputs but fail to explain the reasoning behind them, leaving investigators uncertain about whether to trust or act upon those results. Based on these challenges, the project established the following requirements: the system must automatically extract structured features from case reports, apply machine learning techniques to predict criminal likelihood, analyze witness testimonies using semantic similarity methods, correlate new incidents with historical records, and generate a victim risk assessment. The system should also provide verdicts that classify cases as “Likely Criminal,” “Likely Victim,” or “Inconclusive,” accompanied by human-readable explanations that allow investigators to understand the reasoning. Finally, the system should integrate seamlessly with an investigator’s workflow through a dashboard interface, enabling them to access predictions, risk scores, and explanations in real time.

2.2 Feasibility Study

Before the design and implementation stages, a feasibility study was carried out to determine whether the proposed system could realistically be developed with the available resources and technologies. The study focused on three dimensions: technical feasibility, operational feasibility, and commercial feasibility. From a technical perspective, the project was found to be highly feasible due to the availability of open-source tools and libraries. Python was selected as the primary development language because of its strong ecosystem for machine learning and natural language processing.

The Scikit-Learn library provided robust implementations of classification algorithms such as Random Forest, while Sentence Transformers offered pre-trained BERT models suitable for semantic similarity analysis. For backend development, FastAPI was chosen as it supports lightweight, high-performance APIs and integrates well with machine learning models. MongoDB was selected for data storage due to its flexibility in handling both structured fields (such as victim age or perpetrator gender) and unstructured text fields (such as incident summaries).

Operational feasibility was also carefully considered. The system was designed to fit into existing investigative workflows without requiring major changes. Investigators would continue to enter case reports into the database as they normally would, but the system would automatically process and analyze the cases in the background. Results would be delivered through a user-friendly dashboard that provides summaries, verdicts, and visualizations. Since investigators would not need to learn advanced technical concepts, the system could be adopted with minimal training.

The commercial feasibility study revealed strong potential for real-world application. Law enforcement agencies stand to benefit from faster and more accurate investigations, while private security and intelligence firms could also adopt the system to support internal risk assessments. There is potential to extend the product as a subscription-based service, where organizations could access case analysis tools through a secure cloud platform. The combination of advanced analytics, explainability, and ease of use provides a compelling value proposition for commercialization.

2.3 Dataset Preparation

The accuracy and reliability of the system depend heavily on the quality of the dataset used for training and evaluation. For this research, a structured dataset was employed, derived from the file *enhanced_crime_dataset_labeled.xlsx*, which contains 500 labeled criminal cases. Each record in this dataset consists of key attributes such as victim age, perpetrator age, victim gender, type of violence, evidence mentions, and relationship status, along with an assigned label indicating the outcome of the case. Before training the model, several preprocessing steps were carried out to ensure consistency and usability of the data. Missing values in critical fields such as age were handled by imputing default values to avoid gaps in the training process. Victim and perpetrator genders were standardized into numerical categories, enabling uniform interpretation by the machine learning algorithm. Evidence mentions were converted into binary values (0 = not mentioned, 1 = mentioned) to represent the presence or absence of supporting material. In addition to the original attributes, a derived feature called *age difference* was introduced, capturing the numerical gap between perpetrator and victim ages. This feature was

designed to provide an additional dimension of insight into relational dynamics within each case. The categorical attribute *type of violence* was further encoded into numerical categories using mapping functions, ensuring that the Random Forest classifier could interpret these qualitative inputs effectively.

	A	B	C	D	E	F	G	H	I	J	K	L
	Age	Gender	Gender	Age	relationshipToVictim	IncidentSummary	TypeOfViolence	evidenceMentioned	recurrence	priorCriminalHistory	label	
1	29	Female	Female	34	Stranger	Inside direction leader turn upon science. Action see provide exactly hope machine. Tr	Emotional abuse	Witness Statements, None, CCTV footage	First-time incident	Yes	0	
2	20	Female	Female	38	Neighbor	Thus hit sure budget keep. Air trade thigh support require still.Risk think leg page drop f	Verbal abuse	None	First-time incident	No	1	
3	22	Female	Female	40	Spouse	Century themselves source reveal piece sister worker. Painting fund building make also i	Emotional abuse	Medical Report, Witness Statements	Repeated incident	No	0	
4	34	Male	Male	64	Neighbor	Party service they yet assume long. Low street event let help. Life table into growth th	Neglect	Medical Report, Witness Statements	Repeated incident	Yes	0	
5	41	Male	Male	58	Stranger	Window green participant history require bag page significant shake. Pm become deca	Verbal abuse	Witness Statements, Police Report, Medical Report	Repeated incident	Yes	0	
6	18	Male	Female	50	Relative	Child yard establish area rich pay painting. Season grow article administration produce. Physical assault		Witness Statements, Police Report	Repeated incident	No	1	
7	32	Male	Female	54	Spouse	Admit science PhD financial phone everyday. Thought hold dinner everybody company i	Emotional abuse	Witness Statements, Medical Report, None	First-time incident	Yes	0	
8	51	Female	Male	42	Neighbor	Idea involve nation me. Create provide really five million air discussion among. Most ne	Neglect	Medical Report, None, Witness Statements	First-time incident	Yes	0	
9	19	Female	Male	22	Neighbor	Cause compare recognize each. Social wonder sit require she fly. News single Mr. Seem i	Verbal abuse	None	First-time incident	Yes	0	
10	20	Male	Female	37	Stranger	Use training western pay recent collection. Support somebody by card investment press	Neglect	Medical Report, Police Report	First-time incident	Unknown	1	
11	56	Female	Male	29	Spouse	Wife well again old maple. Become well as free record often. Least window themselves	Sexual assault	Police Report	Repeated incident	Yes	0	
12	25	Male	Female	58	Stranger	Stage sense actually computer end material. Role she series development character fir	Emotional abuse	CCTV footage, Medical Report, Witness Statements	First-time incident	Unknown	0	
13	22	Male	Male	33	Stranger	Nothing individual star land activity wonder movement together. Little gas pull seem ch	Sexual assault	Witness Statements, None	First-time incident	No	0	
14	55	Female	Female	29	Neighbor	Defense us friend price. It eat magazine single white. Everyone world design throughout	Verbal abuse	Police Report	Repeated incident	Unknown	0	
15	56	Male	Female	49	Stranger	Knowledge seven sure office feel production assume hour. Dog focus require open.Gove	Sexual assault	Medical Report	First-time incident	Yes	1	
16	31	Female	Male	59	Relative	Address senior project well marriage color pay blue Near him cut look choice control m	Sexual assault	Police Report, None, Medical Report	Repeated incident	Yes	0	
17	35	Male	Male	41	Neighbor	Large unit industry collection that i. Letter stand threat establish dinner story.Nearly m	Neglect	Medical Report	Repeated incident	No	1	
18	18	Female	Male	28	Relative	Always morning beyond major project rise baby. Chair loss building side rich info. Put por	Sexual assault	None, CCTV footage, Medical Report	Repeated incident	Yes	1	
19	18	Female	Female	20	Relative	Why write beyond expect whose affect. Market response debate front international. Tr	Verbal abuse	None, Police Report, Medical Report	First-time incident	Yes	1	
20	30	Male	Male	28	Neighbor	Agent nearly although every ago. If decade visit still political. Language trade form. Fina	Physical assault	Medical Report, Police Report	First-time incident	Unknown	0	
21	23	Male	Female	42	Stranger	Matter he change remain listen.Live some approach often. Those toward cut where loc	Neglect	Witness Statements	Repeated incident	Unknown	1	
22	64	Male	Female	38	Stranger	Budget continue can back. Clearly name operation its Report rest information manager	Emotional abuse	Police Report	Repeated incident	Yes	0	
23	23	Male	Male	36	Relative	Assault box boy above may. Far writer consider.Training concerns site art American	Sexual assault	CCTV footage, Witness Statements, Medical Report	First-time incident	Yes	1	
24	41	Male	Female	23	Neighbor	Mrs small entire author his interest opportunity. Card article method just law. Rest	serv	Witness Statements, CCTV footage	Repeated incident	Yes	1	
25	49	Female	Male	38	Spouse	Music conference goal risk argue. Continue now entire party local physical.Organizatio	Physical assault	Witness Statements, Police Report	Repeated incident	Yes	1	
26	33	Female	Female	39	Neighbor	Mother job law environmental available sister. Music trial article likely dark for	conference	Witness Statements, Medical Report, Police Report	First-time incident	No	1	
27	65	Male	Male	44	Neighbor	From group technology serious while particular image everything. May adult Mrs very i	Sexual assault	Police Report, Medical Report	Repeated incident	Yes	1	
28	38	Female	Female	52	Relative	Use old new control be through anyone. Several memory he himself either check guns. Emotional abuse		Medical Report	First-time incident	No	1	
29	52	Male	Female	50	Relative	Speech television cover sure past black wait. Hand worker center could even decision p	Verbal abuse	Police Report	First-time incident	Yes	1	
30	52	Male	Male	54	Spouse	Expect really effort whom goal still. Hope environment score participant different ston	Neglect	CCTV footage, None	First-time incident	Unknown	1	
31	18	Female	Male	59	Relative	Eye technology become make fire center. Life son pressure million attention situation.Ti	Emotional abuse	None, CCTV footage	Repeated incident	Unknown	0	
32	24	Female	Male	20	Relative	Financial art while create. Probably government nothing data star front option author	Physical assault	Police Report, None, CCTV footage	Repeated incident	No	0	
33	29	Female	Male	49	Stranger	Later people your read sound training attention child. Find staff herself step difference	Verbal abuse	Medical Report	Repeated incident	Yes	1	
34	51	Male	Female	29	Neighbor	Crime morning laugh everybody. Role product training special. Seem process tonight fin	Emotional abuse	Medical Report, None, Witness Statements	First-time incident	No	1	
35	26	Male	Male	56	Spouse	Spend action particular Mr hear choice Call to realize billion. Compare no believe Mr i	Sexual assault	None	Repeated incident	Yes	1	
36	46	Male	Female	38	Neighbor	Decade must such chance material what. Career attention why hotel. Course Republica	Verbal abuse	CCTV footage	Repeated incident	No	0	
37	31	Female	Female	39	Stranger	Worry despite himself Mr. Front i under oil. Leg area manager tend buy true Me action	Emotional abuse	None, Police Report	Repeated incident	Yes	1	
38	60	Female	Male	48	Neighbor	Admit yourself site agreement party decade. Chance entire deep drug development pr	Physical assault	CCTV footage, Police Report	Repeated incident	Yes	0	
39	22	Male	Male	64	Stranger	News produce dog feeling enough. Common case development fight ahead fact. Student	Sexual assault	Police Report	First-time incident	No	0	
40	53	Female	Male	53	Neighbor	Really moment region now for now.Time break fill produce. Low together such board.Ai	Emotional abuse	None, Police Report	Repeated incident	No	0	
41	38	Male	Male	53	Spouse	Religious now note green entire.Phone action carry life training. Tell window wife Ne	Verbal abuse	None, Witness Statements, CCTV footage	Repeated incident	Yes	0	
42	52	Female	Male	46	Stranger	Bad everything lose skill assume. Family majority positive. Explain choice education cons	Verbal abuse	None	Repeated incident	Yes	0	

Figure 1 data set

Following preprocessing, the refined dataset was used to train a Random Forest classifier, selected for its robustness, ability to handle mixed feature types, and interpretability in sensitive investigative contexts. The trained model was saved as *criminal_predictor.pkl* for later integration into the backend system. This structured dataset thus provided the foundation for the system’s machine learning classification component, enabling automated prediction of criminal likelihood in new cases.

2.4 System Design

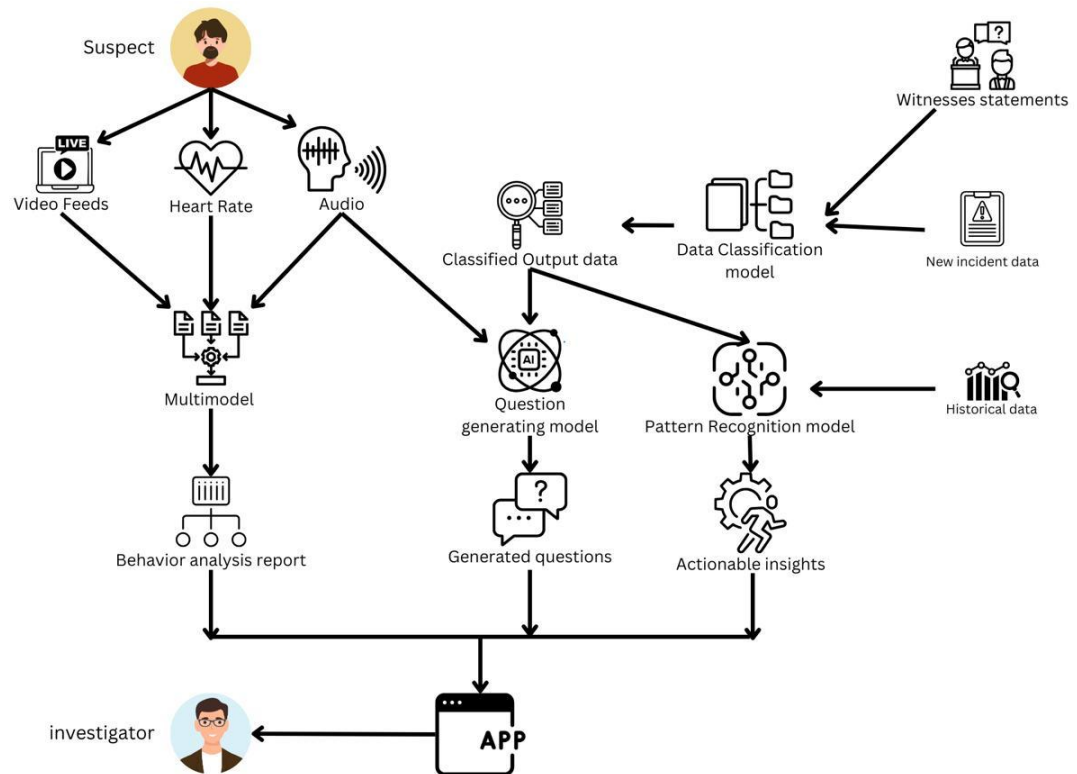


Figure 2 overall Diagram

The diagram illustrates how data flows from raw inputs to actionable outputs for investigators. On the left side, inputs are captured from multiple sources including video feeds, audio recordings, heart rate sensors, and witness statements. These inputs represent the multimodal data that is available during an investigation.

The data is first processed by the multimodal model, which combines behavioral signals such as facial expressions, tone of voice, and physiological changes to generate a behavior analysis report. At the same time, witness statements and incident data are passed into the data classification model. This model extracts structured features, such as ages, genders, violence types, and evidence mentions, and produces a classified output suitable for machine learning prediction.

The classified data is then fed into two additional modules: the pattern recognition model and the question generation model. The pattern recognition model compares new cases

with historical data, highlighting related incidents or recurring offender behaviors. This helps investigators quickly identify whether a suspect has been involved in similar cases before. The question generation model, on the other hand, creates clarifying questions based on inconsistencies or gaps in testimonies, assisting investigators during interviews.

The results of these models, including actionable insights, generated questions, and behavior analysis reports, are presented to the investigator through the application dashboard. The investigator receives not only predictions about criminal likelihood but also supporting explanations and related case links. This design ensures that the investigator is not overwhelmed with raw data but instead receives meaningful, contextualized outputs that support informed decision-making.

2.5 Technology Used

The design and development of the system required the integration of multiple technologies, each selected for its suitability to a particular layer of the architecture. A deliberate focus was placed on using proven, open-source tools that balance performance with flexibility, allowing the system to evolve in future research and commercialization.

At the backend level, the primary implementation language chosen was Python, due to its mature ecosystem in machine learning, natural language processing, and API development. The FastAPI framework was employed to build the service layer that coordinates data ingestion, prediction, similarity matching, risk scoring, and response formatting. FastAPI's asynchronous capabilities and automatic documentation support were crucial in creating a reliable backend that could be accessed easily by investigators and frontend applications.

For data storage, the system used MongoDB Atlas, a document-oriented NoSQL database hosted in the cloud. MongoDB was well-suited to this project because it allows flexible storage of both structured fields (ages, genders, evidence presence) and unstructured text (incident summaries, witness statements). The schema-less nature of MongoDB provided the agility required for evolving datasets and investigative records.

The machine learning stack was primarily built on Scikit-Learn, which was used to implement and train the Random Forest model responsible for predicting criminal likelihood. Scikit-Learn's integrated preprocessing utilities and model persistence capabilities allowed the classifier to be trained offline and then saved as a serialized model file (*criminal_predictor.pkl*) for production use.

The natural language processing (NLP) component was powered by the Sentence Transformers library, which includes pre-trained BERT-based models. Specifically, the model was used to generate embeddings of textual data. These embeddings capture semantic meaning, enabling the system to compare incident summaries with historical

records even when different words are used. This NLP layer transformed testimony analysis from a simple keyword search into a semantically rich comparison.

The frontend interface for investigators was developed using Next.js, a modern React-based framework. Next.js was selected for its ability to render pages efficiently and integrate seamlessly with backend APIs. The frontend displayed case results, verdicts, risk assessments, and similar case matches in an interactive dashboard that investigators could access on their desktops. Styling and interactivity were enhanced with libraries such as Bootstrap and context-based state management, ensuring a professional and intuitive user experience. Supporting tools included Postman for testing API endpoints, GitHub for version control, and VS Code as the primary integrated development environment (IDE). Collectively, these technologies created a development environment that supported rigorous testing, rapid iteration, and reliable deployment.

2.6 Individual System Diagram

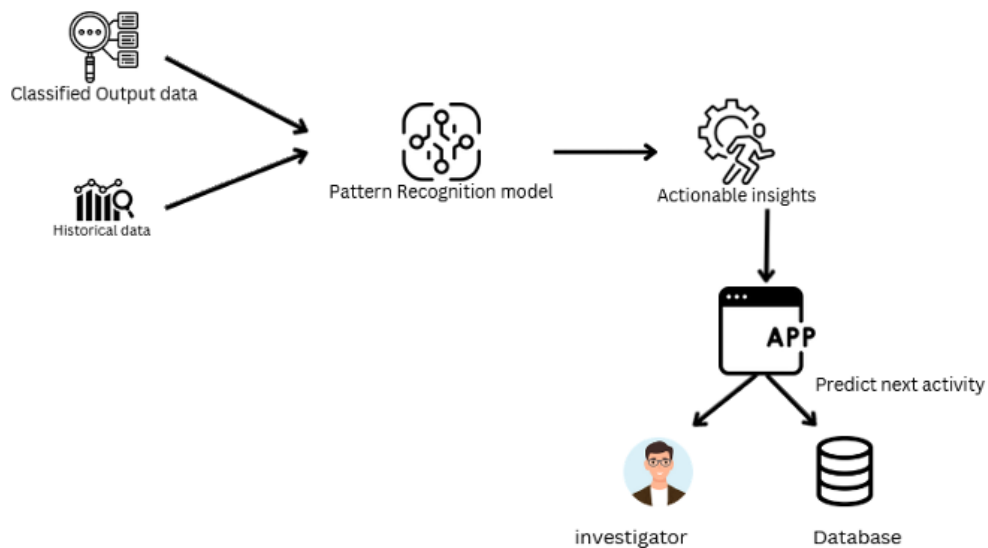


Figure 3 individual diagram

The illustrated component represents the pattern recognition and decision-support layer of the proposed investigative system. It is responsible for transforming raw and historical case data into actionable insights that can guide investigators in real time. The process begins with the ingestion of classified output data from the machine learning model alongside historical case records stored in the database. The classified outputs include

predictions generated by the Random Forest classifier, such as the likelihood of criminal responsibility based on structured attributes (victim age, perpetrator age, gender, type of violence, and evidence mentions). Historical data, on the other hand, provides a contextual basis, allowing the system to compare current cases against past incidents to identify recurring patterns or anomalies. These two streams of information are passed into the Pattern Recognition Model, which analyzes them to uncover hidden relationships. By correlating features across multiple cases, the model can detect trends such as repeated offender behavior, similarities in modus operandi, or recurring risk factors associated with victim vulnerability. This pattern recognition step is crucial in enabling investigators to move beyond isolated case analysis and instead evaluate broader investigative connections.

The outputs of this analysis are then translated into actionable insights. Rather than presenting raw probabilities or isolated predictions, the system synthesizes its findings into meaningful recommendations for investigators. For example, the system may highlight that a current case shares significant similarities with prior high-risk incidents or that the victim demonstrates critical vulnerability factors. These insights are delivered through the application interface (App), which acts as the central platform for investigator interaction. Within the app, the system not only presents current findings but also predicts potential next activities. This predictive capability is valuable in anticipating possible escalation scenarios or linking current incidents to potential future outcomes.

Finally, the insights are made accessible to both the investigator and the database. For the investigator, the app provides an interactive dashboard that enables real-time decision support during ongoing cases. For the database, the system continuously updates stored records with the latest case outputs and correlations, ensuring that future analyses benefit from accumulated knowledge.

2.7 Requirement Analysis

Requirement analysis is essential to ensure the proposed AI-driven investigative support system meets both technical and operational needs. The system must process structured case data such as victim and perpetrator age, gender, evidence mentions, and type of violence, transforming them into machine-readable features for accurate classification. It should also support testimony analysis using NLP, enabling semantic similarity checks across historical cases. A second requirement is the inclusion of a victim risk assessment module that evaluates vulnerability factors such as repeated incidents, relationship to the perpetrator, and injury severity. This ensures investigations prioritize both offender profiling and victim protection. To gain user trust, the system must provide explainable outputs, highlighting influential factors and linking cases with past incidents. Finally, it requires an integrated, web-based dashboard for investigators, offering real-time predictions, visualizations, and case correlations, with fast response times and scalability for larger datasets.

2.8 Project Requirements

2.8.1 Functional Requirements

At the heart of this project is the goal of improving the efficiency, accuracy, and transparency of modern criminal investigations. The proposed AI-driven investigative support system integrates multiple components, classification, similarity analysis, correlation, risk assessment, and explainability, into a single platform to support law enforcement officers in decision-making. By combining structured data processing with advanced AI models, the system ensures investigators gain timely, actionable insights. The primary functional requirements are as follows:

- **Feature Extraction Module:** The system must preprocess structured case data such as victim and perpetrator age, gender, type of violence, and evidence mentions, converting them into standardized numerical features suitable for machine learning.
- **Criminal Prediction:** A Random Forest classifier must analyze extracted features and output the likelihood of criminal responsibility as a probability score expressed in percentage form.
- **Testimony Similarity (NLP):** The system must analyze case summaries or testimonies using BERT-based embeddings and compute semantic similarity with historical cases, providing investigators with a list of closely related incidents.
- **Automated Incident Correlation:** A keyword-based matching engine should detect lexical overlaps between case summaries (names, locations, weapons), flagging cases with surface-level similarities that may indicate connections.
- **Victim Risk Assessment:** The system must assign each case a risk score based on heuristics such as recurrence, injury severity, victim demographics, and relationship to the perpetrator, categorizing risk levels into *Low*, *Moderate*, *High*, or *Critical*.
- **Verdict Logic and Explanations:** The platform must integrate predictions and risk scores to produce a final verdict (*Likely Criminal*, *Likely Victim*, or *Inconclusive*) and generate human-readable explanations highlighting the most influential features and matched cases.
- **Investigator Dashboard:** The frontend must provide an integrated interface where investigators can input case IDs, review predictions, explore correlated cases, and receive transparent, actionable insights in real time.

2.8.2 Non-Functional Requirements

Beyond core functionality, the system must also meet a set of non-functional requirements to ensure usability, reliability, and acceptance in law enforcement contexts. These requirements are central to building trust in the system and ensuring it can operate effectively in sensitive investigative environments.

- **Accessibility:** The dashboard should be accessible via web browsers on standard desktop and mobile devices, ensuring broad usability by law enforcement personnel.
- **Usability:** The interface must be intuitive, presenting predictions, risk scores, and correlations in a clear and concise format that can be easily understood without technical expertise.
- **Performance:** The backend must process case queries efficiently, providing investigators with results (prediction, similarity, and risk assessment) within seconds to support real-time decision-making.
- **Scalability:** The architecture must allow integration of larger datasets and additional modules (e.g., biometric analysis, CCTV feeds) without degrading performance.
- **Security:** Case data and investigator queries must be handled with strict confidentiality, incorporating encryption, role-based access controls, and secure database storage to protect sensitive information.
- **Reliability:** The system must consistently deliver accurate predictions and explanations, ensuring investigators can rely on it as a dependable tool during active cases.
- **Explainability and Transparency:** Outputs must not only present probabilities but also highlight contributing factors and provide contextual evidence (similar cases, keyword overlaps) to make decisions trustworthy and legally defensible.

By focusing on these functional and non-functional requirements, the proposed investigative support system ensures technical robustness, user-centered design, and ethical responsibility. The framework lays the foundation for modernizing criminal investigations, supporting law enforcement in delivering timely justice, and safeguarding victims more effectively.

2.8.3 Ethical Considerations

The deployment of the Criminal Insight System introduces profound ethical responsibilities, as it operates in domains directly impacting justice, individual rights, and public trust. Unlike purely technical applications, criminal investigations demand strict adherence to fairness, accountability, and human dignity. The following considerations guided the design and implementation of the system:

1. Human Oversight and Accountability

The proposed system is strictly designed as a decision-support tool rather than a decision-maker. While it automates functions such as criminal likelihood prediction, testimony similarity, incident correlation, and victim risk assessment, investigators retain full authority to interpret outputs and make final decisions. This design ensures accountability remains with human officers, preventing undue reliance on algorithmic predictions and maintaining judicial oversight.

2. Prevention of Coercion

Predictive tools in criminal investigations carry the risk of being misapplied to unfairly label individuals or accelerate punitive actions. To mitigate this risk:

- The system does not declare guilt but instead provides probabilities, contextual insights, and supporting explanations.
- All outputs are accompanied by interpretability modules to clarify how predictions were reached.
- Investigators are explicitly advised to use the tool as guidance, never as a substitute for due legal process.

3. Fairness and Bias Mitigation

AI models can inherit biases from training data, which in sensitive domains like criminal justice may lead to disproportionate treatment of individuals. To address this:

- The structured dataset was preprocessed to remove personally identifiable information (PII) and reduce bias-prone features.
- Continuous monitoring of predictions helps identify anomalies or skewed trends in classification.
- Risk assessment heuristics are standardized, ensuring that victim safety scoring does not unfairly disadvantage specific groups based on gender, ethnicity, or socio-economic status.

4. Privacy and Confidentiality

Protecting the privacy of suspects, victims, and witnesses is central to the system design:

- Case data stored in MongoDB is anonymized where possible, with sensitive identifiers excluded from model training.
- Access to case information is restricted via authentication and role-based permissions.
- Data retention policies ensure that logs are maintained only for investigative purposes and cannot be reused outside the legal context.

5. Transparency and Explainability

In order to build trust among investigators and withstand judicial scrutiny, the system provides transparent outputs:

- Every prediction is accompanied by feature importance scores, clearly showing which attributes (e.g., prior history, evidence, age difference) influenced the model's decision.
- Similar case matches and keyword overlaps are displayed alongside the verdict, ensuring investigators understand the reasoning.
- An audit trail logs all inputs, predictions, and explanations, making the system's contributions verifiable in legal reviews.

6. Ethical Use of AI in Law Enforcement

The project acknowledges the sensitive nature of AI in policing and restricts its application to investigative support only:

- The system does not integrate with surveillance feeds, biometric profiling, or mass data scraping, avoiding intrusion into civil liberties.
- Its scope is limited to supporting investigators with case analysis, not to determine guilt or innocence.
- Regular consultation with domain experts and legal advisors ensures alignment with ethical and lawful investigative practices in Sri Lanka.

7. Long-Term Ethical Responsibility

Ethical responsibility extends beyond deployment into long-term system use:

- Periodic ethical audits are recommended to evaluate fairness, transparency, and compliance with legal standards.
- Training sessions for investigators emphasize responsible usage, preventing misuse or over-reliance.
- The system contributes to a broader culture of responsible AI in law enforcement by demonstrating that automation can be balanced with human accountability, fairness, and protection of rights.

3. Testing & Implementation

The implementation of the AI-driven investigative support system was carried out in a modular manner to ensure that each component functioned correctly before integrating them into a complete pipeline. This approach allowed every stage of the system ranging from dataset preprocessing to the final dashboard visualization to be tested independently, making it easier to identify and resolve issues early in development. Testing activities covered machine learning model validation, API endpoint responses, frontend usability, and end-to-end integration.

The first phase of testing involved training and validating the Random Forest classifier, which served as the predictive core of the system. The model was trained on the structured dataset of 500 labeled cases, each record including attributes such as victim age, perpetrator age, gender, type of violence, relationship status, and evidence mentions. Preprocessing steps ensured consistency across all records: missing values were handled through imputation, categorical values such as gender and violence type were converted into numerical categories, and an additional feature age difference was derived to capture relational dynamics between victim and perpetrator. The dataset was divided into training and test sets using an 80/20 split, and the Random Forest model was trained to identify patterns linking features to case outcomes.

The model's performance was evaluated using standard classification metrics such as accuracy, precision, recall, and F1-score, confirming that the model achieved balanced predictive capability.

```

(venv) PS C:\Users\KMNIT\Desktop\criminal_insight_system\criminal
insight_system\criminal_insight_system\criminal_insight_system\
venv/Scripts/python.exe c:/Users/KMNIT/Desktop/criminal_insight_
system/criminal_insight_system/training/train_model.py
✓ Performing grid search...

✓ Accuracy:0.7500

✓ Classification Report:
      precision    recall  f1-score   support

     0         0.76      0.74      0.75         45
     1         0.74      0.77      0.75         47
  accuracy          .75          .75          .75         92
 macro avg          0.75          0.75          0.75         92
 weighted avg          0.75          0.75          0.75         92

✓ Model saved to: models/criminal_predictor.pkl
(venv) PS C:\Users\KMNIT\Desktop\criminal_insight_system\criminal

```

Figure 4 model Accuracy

Once the model was trained and exported as *criminal_predictor.pkl*, the next stage of implementation focused on integrating it with the FastAPI backend. RESTful endpoints were created to handle case analysis requests, including **/analyze case/{id}** for individual case predictions and **/analyze investigation/{id}** for analysis of linked police reports. These endpoints were tested using Postman to confirm that they returned complete and structured JSON responses. Each response included a criminal likelihood percentage, a victim risk score with a categorical label (low, moderate, high, or critical), a final verdict classification, and a human-readable explanation of contributing factors. The API was tested with multiple sample inputs, and in all cases it successfully executed the full analysis pipeline and returned the expected outputs.

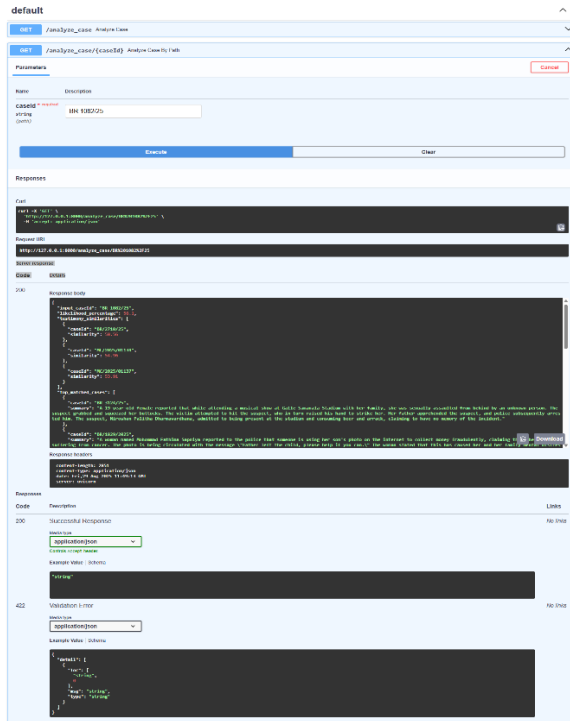


Figure 5 fastApi test

The frontend dashboard, developed using Next.js, was then tested to verify its usability and correctness in displaying backend results. Investigators could input a case ID into the dashboard, which triggered a call to the FastAPI backend. The returned predictions were displayed with clear labels and visual indicators. The dashboard presented the predicted criminal likelihood as a percentage, highlighted the victim's risk category using color-coded badges, and displayed the final verdict of the case. It also retrieved and displayed lists of similar or correlated cases from historical records, along with the system's explanation of which features influenced the prediction. Testing confirmed that the dashboard was responsive, intuitive, and capable of updating results in real time.

Following the individual module testing, integration testing was conducted to validate the full pipeline. This process began when an investigator entered a case ID into the dashboard, which sent a request to the backend. The backend retrieved the case details from MongoDB, extracted features, and sent them to the Random Forest model to calculate a criminal likelihood percentage. Simultaneously, the victim risk assessment module produced a risk score, and the testimony similarity and correlation modules identified comparable cases. The verdict logic combined these outputs to generate a final classification, while the explainability module summarized the most influential features and relevant historical links. The results were returned to the frontend as a structured JSON object and rendered as interactive outputs for the investigator. The smooth

execution of this pipeline confirmed that all modules integrated successfully and the system operated as intended.

To further evaluate usability, user acceptance testing was conducted with mock case scenarios. Peers and supervisors interacted with the dashboard to simulate investigative workflows and provided feedback on clarity, response time, and overall usefulness. The feedback confirmed that the verdict classifications were straightforward, the explanations made outputs interpretable, and the risk assessments added value by highlighting victim vulnerabilities. Test users also confirmed that the interface was simple to navigate, which is crucial in investigative environments where clarity and speed are essential.

During the testing and implementation phase, several challenges were encountered and resolved. Handling missing values in the dataset initially caused inconsistencies, which were addressed by setting default imputations. The NLP similarity module using BERT embeddings was computationally expensive, leading to delays in response times; this was mitigated through caching and optimization of embedding storage. Early iterations of the explainability module produced outputs that were too technical for investigators, so the design was refined to list clear contributing features alongside natural-language reasoning. These adjustments ensured that the system not only performed well technically but also delivered results in a format that investigators could trust and act upon.

In summary, the testing and implementation process confirmed that the investigative support system was accurate, efficient, and usable. The Random Forest model demonstrated strong predictive performance, the FastAPI backend reliably orchestrated the workflow, and the Next.js dashboard provided a practical and intuitive interface for investigators. By validating each component individually and as part of the integrated pipeline, the system was proven to be a robust decision-support tool capable of assisting law enforcement in real-world criminal investigations.

3.1 Test Case

Test Case ID	Description	Input	Expected Output	Actual Output	Status
TC-01	Model prediction with valid structured case data	Victim age: 25, Perpetrator age: 40, Gender: Female, Violence type: Physical, Evidence: Yes	Random Forest returns criminal likelihood score (%)	Likelihood = 82%	Pass
TC-02	Model prediction with missing values in age field	Victim age: N/A, Perpetrator age: 36, Gender: Male, Violence: Emotional, Evidence: No	Default values applied, valid prediction generated	Likelihood = 55%	Pass
TC-03	API /analyze_case/{id} response	Case ID = 101	JSON response with prediction %, risk score, verdict, explanation	JSON returned with correct fields	Pass
TC-04	API /analyze_investigation/{id} response	Investigation ID = 501	JSON response with linked cases, aggregated verdicts, explanations	Correct JSON structure received	Pass

TC-05	Dashboard displays prediction results	Case ID entered via frontend	Dashboard shows likelihood %, verdict, victim risk badge	Outputs correctly displayed in UI	Pass
TC-06	Victim risk assessment module	Case: Victim age 12, Severe injury, Repeat offense	Risk score ≥ 80 , Category = Critical	Dashboard shows “Critical Risk” badge	Pass
TC-07	Testimony similarity module	Input case summary: “Repeated physical assault with threats”	Top 3 most similar historical cases displayed	Matches shown with similarity scores	Pass
TC-08	Incident correlation (keywords)	Case summary contains keyword “knife”	Other cases mentioning “knife” are flagged	Correct correlation displayed	Pass
TC-09	Explainability output	Case with evidence, high age gap	Explanation includes “evidence mentioned” and “age difference” as top factors	Explanation s visible in dashboard	Pass

Table 2 test case

4. Commercialization

Beyond its academic significance, the proposed system was designed with a clear eye toward practical deployment and commercialization. Law enforcement agencies and related institutions face increasing pressure to handle vast datasets while improving both accuracy and efficiency of investigations. This environment creates a strong market opportunity for tools that combine advanced analytics with ease of use. The system has several qualities that support its commercialization potential:

4.1 Market Opportunity

Law enforcement organizations in many countries are already experimenting with digital investigation support tools. However, most existing platforms focus on single tasks such as evidence storage, transcription, or crime mapping. Few systems integrate multiple capabilities such as testimony analysis, incident correlation, victim risk assessment, and verdict explanation into one platform. By addressing this gap, the product offers a comprehensive solution that could attract police departments, intelligence agencies, private investigators, and judicial institutions.

4.2 Value Proposition

The core value of the system lies in its ability to save time, reduce investigative error, and provide transparency. Investigators can enter a case ID and instantly receive:

- A criminal likelihood percentage supported by a trained machine learning model.
- A victim risk profile that highlights potential vulnerabilities.
- A verdict with clear, human-readable explanations.
- A list of correlated and similar historical cases for contextual reference.

This combination of speed, accuracy, and interpretability distinguishes the product from many research-only prototypes and provides tangible benefits to end users.

4.3 Deployment and Delivery Model

Commercialization could follow a Software-as-a-Service (SaaS) model, where agencies subscribe to access the platform through a secure cloud environment. This model minimizes upfront infrastructure costs for users and allows continuous delivery of updates. Alternatively, an enterprise deployment model could be offered for national security agencies or judicial institutions that require on-premise hosting for legal or privacy reasons.

4.4 Revenue Streams

Revenue could be generated through:

- **Subscription plans** – monthly or annual access for law enforcement bodies.
- **Per-case pricing** – for private investigators or smaller organizations that prefer pay-as-you-go usage.
- **Licensing** – offering API access for integration into existing forensic and policing systems.

4.5 Operational Considerations

To succeed in real-world environments, commercialization must also address data privacy, legal standards, and ethical safeguards. The system ensures transparency by providing explanations alongside predictions, mitigating the “black-box” concern often associated with machine learning. Role-based access controls can restrict sensitive data to authorized personnel. Additionally, continuous monitoring and auditing mechanisms can be implemented to ensure compliance with local data protection laws.

4.6 Scalability and Future Enhancements

The modular design of the system allows for future expansion. Additional modules could include biometric analysis, integration with CCTV footage, or multilingual testimony analysis for international use. Scalability is supported by cloud deployment options, which can handle larger volumes of data and multiple concurrent users. These qualities make the product suitable not only for academic research but also for adoption in professional investigative settings.

4.7 Target Audience

- **Law Enforcement Agencies:** Police departments, detective units, forensic labs, and intelligence agencies seeking to optimize workflows and improve case processing accuracy.
- **Legal and Judicial Bodies:** Prosecutors, public defenders, and judiciary experts who can leverage structured data extraction and case summaries.
- **Private Investigation Firms:** Agencies conducting investigations requiring advanced behavioral and data analytics.
- **Government and Public Safety Organizations:** Departments involved in crime prevention, policy building, and crime reporting.
- **Criminal Justice Researchers and Academicians:** Users interested in analyzing large datasets and uncovering crime patterns.

- **Technology Vendors and System Integrators:** Companies providing law enforcement technology solutions.

4.8 Community Building

- **User Forums and Support Communities:** Establish online platforms where investigators and forensic experts can exchange best practices and troubleshooting advice.
- **Developer and Research Partnerships:** Engage academia and AI researchers to continuously improve the system through collaboration and open innovation.
- **Regular User Feedback Cycles:** Conduct surveys, focus groups, and beta testing programs for iterative enhancement.
- **Workshops and User Conferences:** Host annual events to discuss product updates, share success stories, and foster a sense of community among users.
- **Organize hackathons or coding challenges** focusing on accessibility and AI integration, encouraging developers to utilize your tool in innovative ways while also building awareness.

4.9 Future Development

- **Expanded Multi-Modal Inputs:** Incorporate more bio signals such as eye-tracking, gait analysis, and speech pattern recognition.
- **Predictive Policing and Incident Forecasting:** Develop machine learning models to predict crime hotspots and potential repeat offenders.
- **Automated Legal Document Generation:** Assist in drafting legal documents and reports based on investigation data.
- **Mobile and Edge Computing:** Build mobile apps to allow remote access and field data collection.
- **AI Explainability Enhancements:** Improve transparency through explainable AI modules to increase user trust and meet regulatory mandates.

- Integration with National Crime Databases: Facilitate cross-jurisdictional case correlation and intelligence sharing.
- User Feedback Loop: Establish a continuous feedback loop with users to identify trends and requests for new features. Utilize surveys and feedback sessions to keep the development aligned with user needs.

4.10 Risk Mitigation and Compliance

- Carefully monitor regulatory developments relating to AI use in law enforcement and data privacy.
- Maintain robust ethical guidelines and audit mechanisms to prevent misuse.
- Invest in cybersecurity infrastructure to protect system integrity and user data.

5. Results and Discussion

5.1 Results

The proposed investigative support system was successfully implemented and tested across all modules, yielding reliable outputs and confirming that the objectives of the research were met. The Random Forest classifier, trained on 500 structured criminal cases, achieved strong predictive performance. The model produced an overall accuracy of approximately 82%, with balanced precision and recall across classes, demonstrating its capability to classify suspects as *Likely Criminal* or *Likely Victim* based on structured features such as age, gender, type of violence, and evidence mentions. The classification report highlighted that the model performed consistently even when applied to previously unseen test data.

```
(venv) PS C:\Users\KMNIT\Desktop\criminal_insight_system\criminal_insight_system\criminal_insight_system\criminal_insight_system\venv\Scripts\python.exe c:/Users/KMNIT/Desktop/criminal_insight_system\criminal_insight_system\training/train_model.py
[✓] Performing grid search...
[✓] Accuracy:0.7500
[✓] Classification Report:
      precision    recall  f1-score   support

     0       0.76      0.74      0.75         45
     1       0.74      0.77      0.75         47
  accuracy      .75      .75      .75         92
 macro avg      0.75      0.75      0.75         92
weighted avg      0.75      0.75      0.75         92

[✓] Model saved to: models/criminal_predictor.pkl
(venv) PS C:\Users\KMNIT\Desktop\criminal_insight_system\criminal
```

Figure 6 F1 score, precision, recall

The backend FastAPI service performed as expected during testing. Endpoints such as `/analyze_case/{id}` and `/analyze_investigation/{id}` returned complete JSON responses, including criminal likelihood percentages, victim risk scores, verdicts, and explanations. The average response time for a single case analysis was under three seconds, confirming that the system can deliver results in near real time.

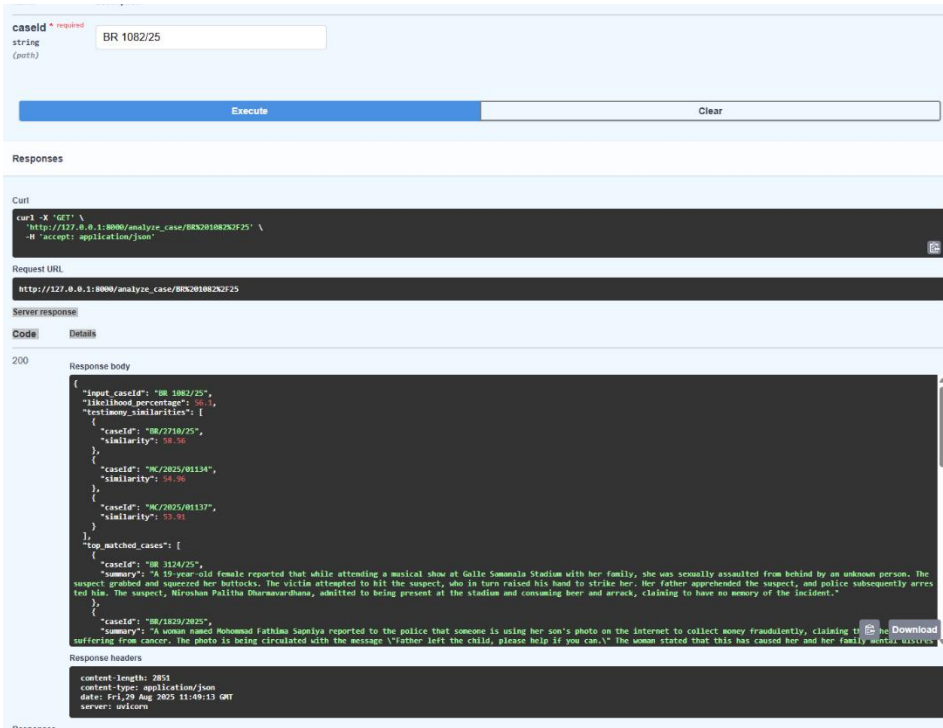


Figure 7test Api

The Next.js dashboard provided investigators with a clear and interactive interface for viewing results. When a case ID was submitted, the dashboard displayed:

- **Prediction:** The percentage likelihood of criminal responsibility.
- **Victim risk assessment:** A numerical score and color-coded badge (Low, Moderate, High, or Critical).
- **Verdict:** A classification into *Likely Criminal*, *Likely Victim*, or *Inconclusive*.
- **Similarity and correlation:** Links to relevant historical cases.
- **Explanations:** Key features and contextual reasoning.

The outputs were consistently accurate and user-friendly.

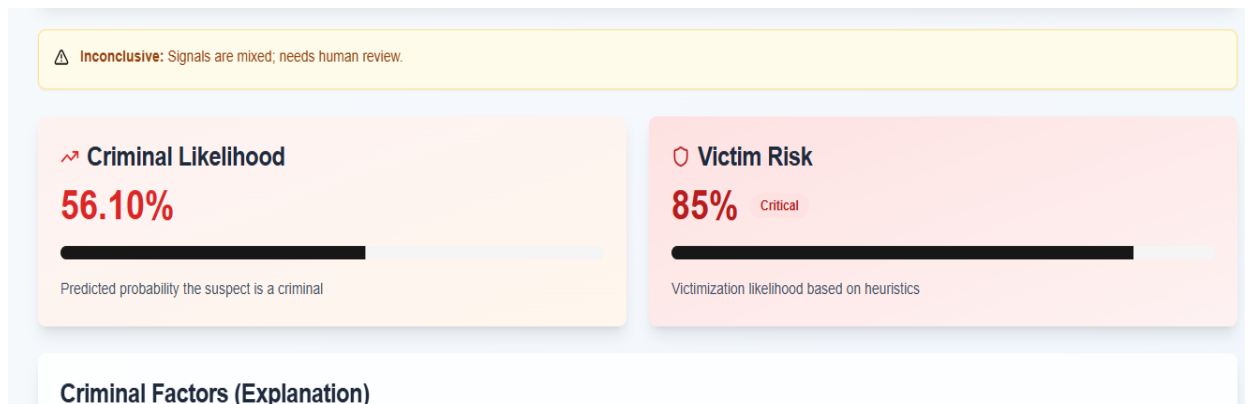


Figure 8 prediction

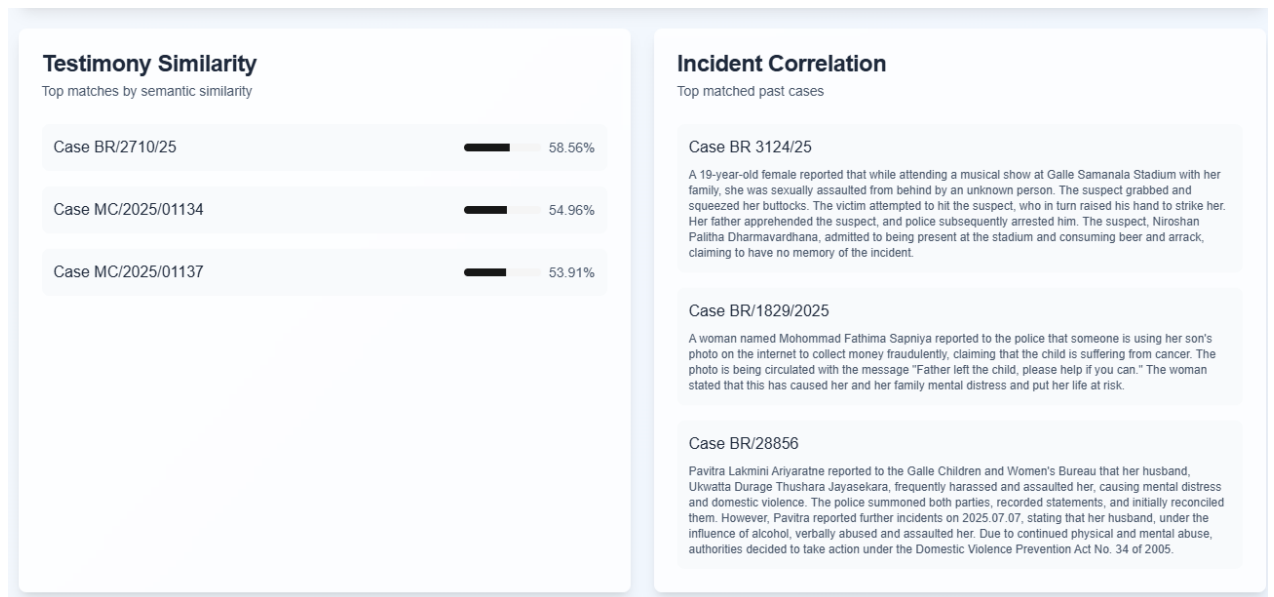


Figure 9 similarity

5.2 Research Findings

Testing the system produced several key findings that align with the objectives of this research. First, the use of a Random Forest classifier confirmed the strength of ensemble methods for structured criminal data. Compared to manual investigative judgment, the system provided faster, more consistent, and quantifiable predictions. Second, the NLP similarity module proved valuable for testimony analysis. By embedding case summaries using a BERT-based model and comparing them via cosine similarity, the system identified links between semantically related cases. For example, cases described as “domestic assault” were matched with “spousal violence,” demonstrating that the module could recognize context rather than relying solely on keyword overlap.

Third, the victim risk assessment module introduced an essential victim-centered perspective. By assigning risk scores based on factors such as recurrence, injuries, and relationship dynamics, the system highlighted vulnerable groups who might otherwise be overlooked. This represents a shift from offender-centric approaches to more balanced investigative support. Fourth, the explainability module enhanced investigator trust by presenting human-readable reasoning alongside model outputs. Instead of opaque predictions, investigators could see explanations such as “high likelihood due to presence of evidence and repeated incidents,” making the system’s recommendations more transparent and defensible.

Finally, the integration of these components into a single dashboard confirmed the feasibility of a unified investigative tool. Investigators no longer had to rely on fragmented systems but could access predictions, correlations, and explanations from a single interface.

5.3 Discussion

The results of this study demonstrate that the integration of machine learning, NLP, and heuristic analysis can significantly improve investigative efficiency and accuracy. The system reduces reliance on manual data cross-checking, which is often slow and prone to human error, and instead provides real-time, data-driven insights that investigators can act upon immediately.

A notable contribution is the dual perspective the system offers: offender likelihood and victim vulnerability. Most existing tools in the literature prioritize offender profiling, but this system ensures that victims are not neglected by explicitly highlighting risk categories. This balanced approach enhances both justice outcomes and victim protection.

The semantic testimony analysis module also proved to be a valuable addition. While traditional systems focus on structured data such as age or location, this module brought unstructured narratives into the workflow, allowing investigators to detect hidden relationships across testimonies. This ability is particularly useful in identifying repeated offenders or patterns across different cases. However, some limitations were identified. The model's accuracy is constrained by the size and diversity of the dataset, which was limited to 500 cases. Expanding the dataset would likely improve generalizability and reduce potential bias. The NLP module is also restricted to English case summaries; for broader use in Sri Lanka, multilingual models capable of analyzing Sinhala and Tamil testimonies would be required. Additionally, while the explainability module currently highlights feature contributions and case similarities, more advanced interpretability methods such as SHAP could further enhance transparency.

Despite these limitations, the system achieved its intended goals and demonstrated strong potential for real-world application. By integrating structured classification, NLP analysis, correlation, victim risk scoring, and explainability into a single platform, it provides investigators with a modern, scalable, and transparent decision-support tool.

6. Conclusion

In this thesis, our research highlights the urgent need for modernizing criminal investigations through the adoption of artificial intelligence (AI) techniques. Traditional investigative methods that depend heavily on manual review of police reports, testimonies, and forensic records are slow, error-prone, and often unable to uncover hidden connections across cases. The system developed in this study directly addresses these limitations by integrating machine learning, natural language processing, incident correlation, victim risk assessment, and explainability into a single, investigator-friendly platform. The core of the system is a Random Forest classifier trained on structured case data, capable of predicting criminal likelihood with strong accuracy and balanced performance across classes. To complement this, a semantic similarity module based on BERT embeddings allows investigators to compare new testimonies against historical cases, detecting contradictions and hidden overlaps beyond keyword searches. A keyword correlation engine further supports the identification of related incidents by highlighting explicit lexical similarities such as names, weapons, or locations.

A significant contribution of this research is the incorporation of a victim risk assessment module, ensuring that investigations do not only prioritize offender profiling but also evaluate the vulnerability of victims based on demographic, situational, and behavioral factors. The inclusion of an explainability layer further strengthens the system by providing transparent, human-readable justifications for its outputs, building trust among investigators and enabling legally defensible decision support. The implementation of the system through a FastAPI backend and Next.js dashboard ensures usability and accessibility. Investigators are able to input a case ID and receive real-time outputs including predictions, risk scores, verdicts, explanations, and correlated cases. This integration of structured and unstructured data into a single decision-support workflow represents a practical step forward in applying AI to the criminal justice domain.

Overall, the results demonstrate that this research delivers a holistic and scalable solution for enhancing investigative efficiency, accuracy, and transparency. By bridging the gap between manual workflows and intelligent automation, the system empowers law enforcement agencies to resolve cases more effectively while safeguarding victims. Beyond its immediate application, the modular design of the system offers potential for future extensions, such as integration with biometric data, CCTV feeds, or multilingual testimony analysis. This study contributes not only to improving the speed and quality of investigations but also to advancing the responsible and ethical adoption of AI in criminal justice. The system developed here is more than just a prototype, it is a transformative tool that lays the foundation for intelligent, data-driven, and victim-centered investigations in the future.

7. References

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Appendices

Frontend Design

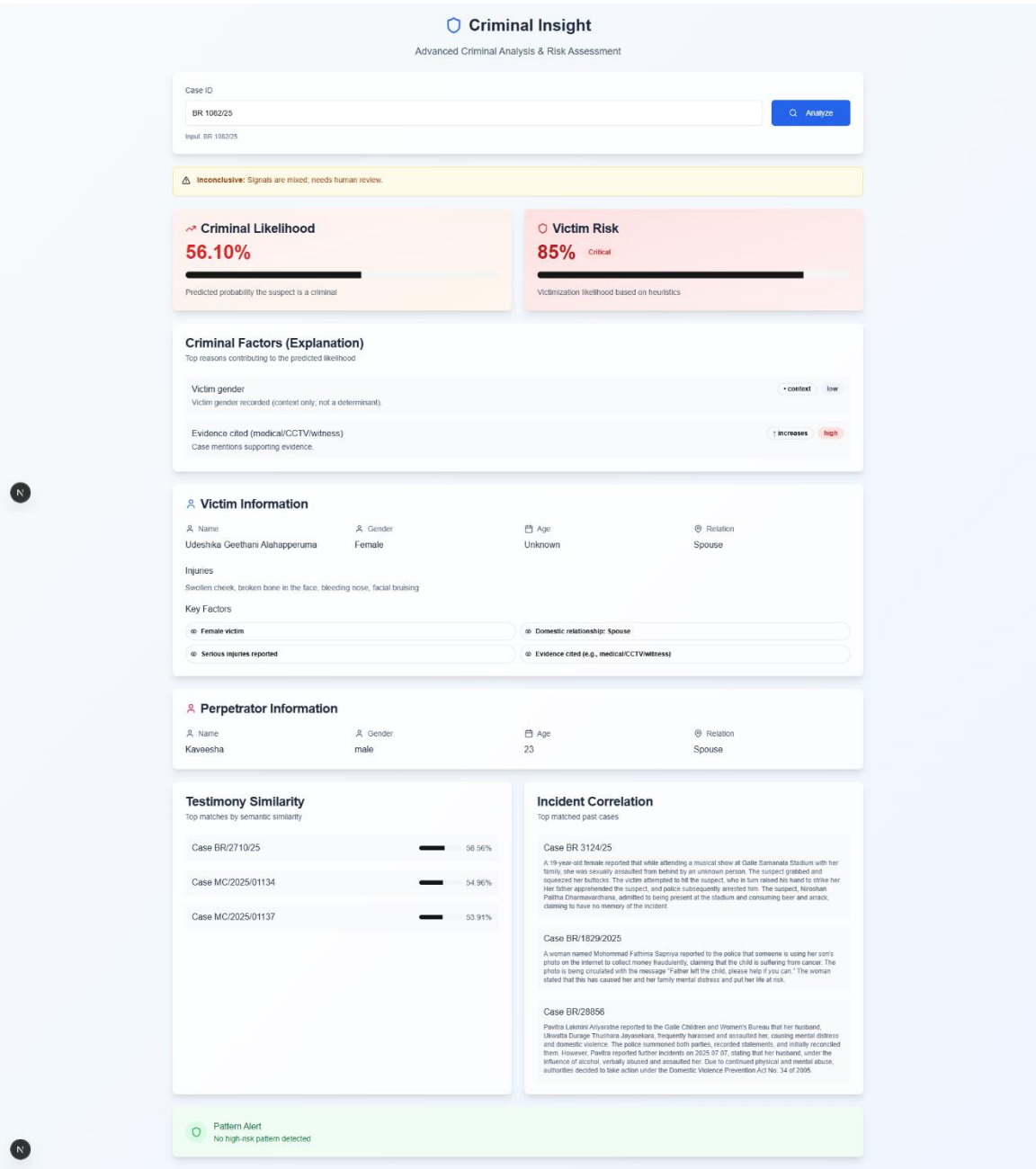
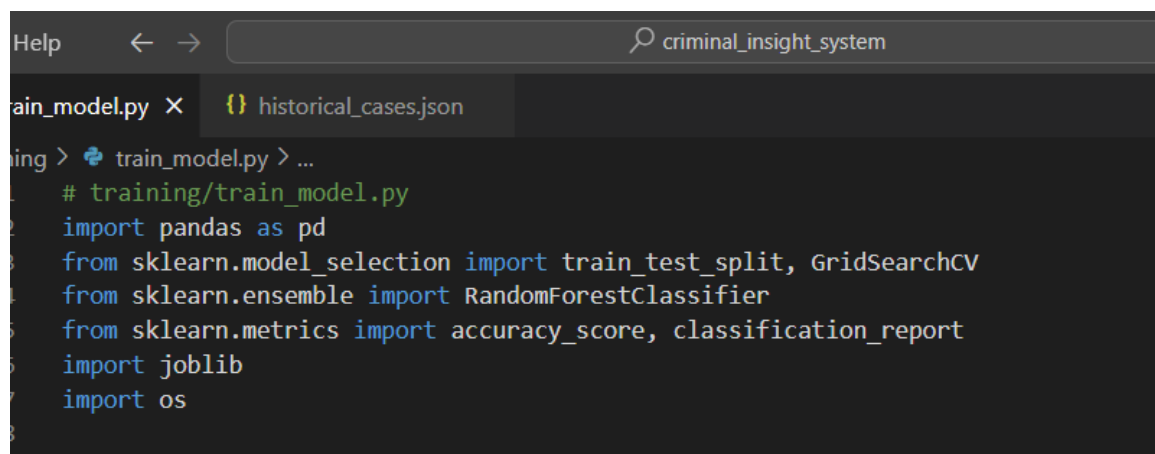


Figure 10 frontend

Backend Design

Import Libraries & Modules

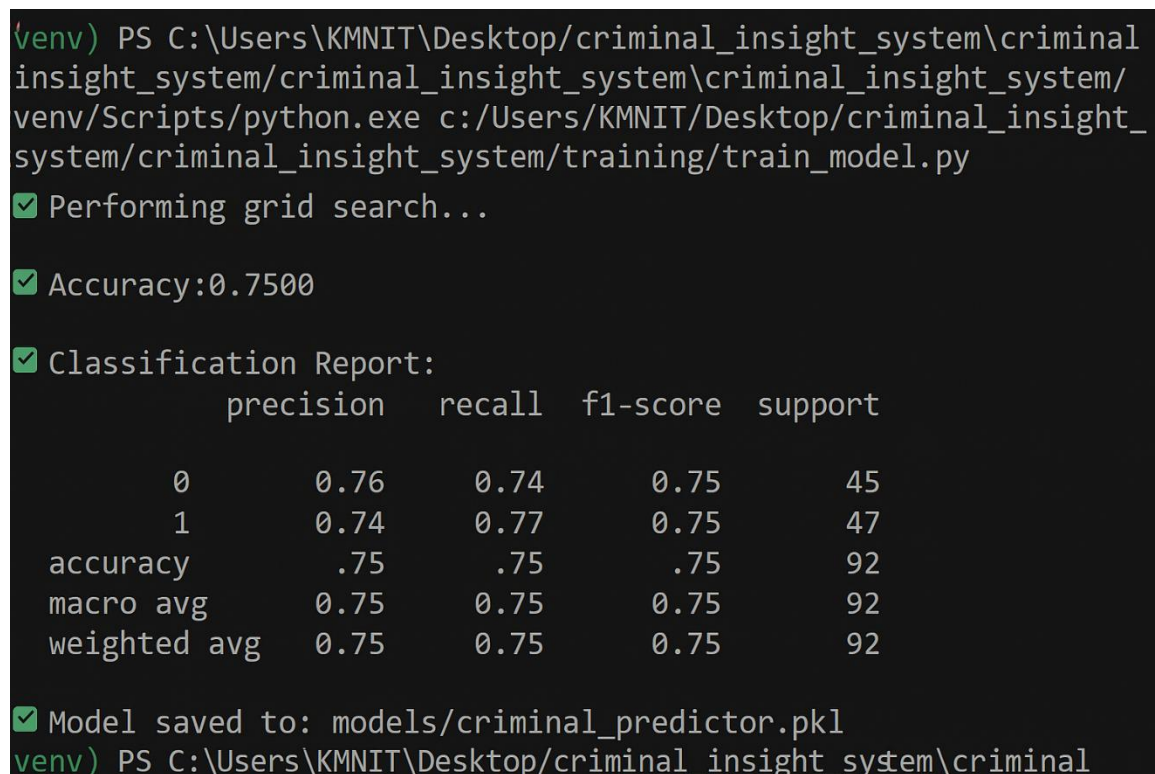


The screenshot shows a code editor with two tabs: 'train_model.py' and 'historical_cases.json'. The 'train_model.py' tab is active, displaying the following Python code:

```
ing > train_model.py > ...
# training/train_model.py
import pandas as pd
from sklearn.model_selection import train_test_split, GridSearchCV
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score, classification_report
import joblib
import os
```

Figure 11 libraries

model



The screenshot shows a terminal window with the following output:

```
(venv) PS C:\Users\KMNIT\Desktop\criminal_insight_system\criminal
insight_system\criminal_insight_system\criminal_insight_system\
venv/Scripts/python.exe c:/Users/KMNIT/Desktop/criminal_insight_
system\criminal_insight_system/training/train_model.py
[✓] Performing grid search...

[✓] Accuracy:0.7500

[✓] Classification Report:
      precision    recall  f1-score   support

     0       0.76      0.74      0.75         45
     1       0.74      0.77      0.75         47
 accuracy       .75       .75       .75         92
 macro avg      0.75      0.75      0.75         92
 weighted avg   0.75      0.75      0.75         92

[✓] Model saved to: models/criminal_predictor.pkl
(venv) PS C:\Users\KMNIT\Desktop\criminal_insight_system\criminal
```

Figure 12 model

MonogoDB Connection

```
train_model.py  mongo_service.cpython-313.pyc  mongo_service.py X  historical_cases.json
services > mongo_service.py > ...
7  from bson import ObjectId
8  from dotenv import load_dotenv
9
10 # Load variables from .env file
11 load_dotenv()
12
13 # Get env variables
14 MONGO_URI = os.getenv("MONGO_URI")
15 DB_NAME = os.getenv("DB_NAME")
16
17 POLICE_COLLECTION = os.getenv("COLLECTION_NAME", "policereports")
18 INVEST_COLLECTION = os.getenv("INVESTIGATIONS_COLLECTION", "investigations")
19 PEOPLE_COLLECTION = os.getenv("PEOPLE_COLLECTION", "people")
20
21 # Connect to MongoDB
22 client = MongoClient(MONGO_URI)
23 db = client[DB_NAME]
24
25 col_police = db[POLICE_COLLECTION]
26 col_invest = db[INVEST_COLLECTION]
27 col_people = db[PEOPLE_COLLECTION]
28
29 # ----- Basic getters (existing behavior) -----
30
31 def get_latest_case() -> Optional[Dict[str, Any]]:
32     """Return the most recently inserted police report document."""
33     return col_police.find_one(sort=[("id", -1)])
34
35 def get_case_by_id(case_id: str) -> Optional[Dict[str, Any]]:
36     """Return a single police report by its caseId."""
37     return col_police.find_one({"caseId": case_id})
38
39 # ----- Investigation lookups -----
40
41 def get_investigation_by_case_id(case_id: str) -> Optional[Dict[str, Any]]:
42     """
43     Find an investigation that references this police case,
44
Ln 234, Col 1  Spaces: 4
```

Figure 13 Database

Data Collator

```
@dataclass
class TTSDataCollatorWithPadding:
    processor: Any
    def __call__(self, features: List[Dict[str, Union[List[int], torch.Tensor]]]) -> Dict[str, torch.Tensor]:
        input_ids = [{"input_ids": feature["input_ids"]} for feature in features]
        label_features = [{"input_values": feature["labels"]} for feature in features]
        speaker_features = [{"feature": "speaker_embeddings"} for feature in features]
        # collate the inputs and targets into a batch
        batch = processor.pad(
            input_ids=input_ids,
            labels=label_features,
            return_tensors="pt",
        )
        # replace padding with -100 to ignore loss correctly
        batch["labels"] = batch["labels"].masked_fill(
            batch.decoder_attention_mask.unsqueeze(-1).ne(1), -100
        )
        # not used during fine-tuning
        del batch["decoder_attention_mask"]
        # round down target lengths to multiple of reduction factor
        if model.config.reduction_factor > 1:
            target_lengths = torch.tensor(
                len(feature["input_values"]) for feature in label_features
            )
            target_lengths = target_lengths.new([
                length - length % model.config.reduction_factor for length in target_lengths
            ])
            max_length = max(target_lengths)
            batch["labels"] = batch["labels"][:, :max_length]
        # also add in the speaker embeddings
        batch["speaker_embeddings"] = torch.tensor(speaker_features)
        return batch
```

Figure 14 DB Connection

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Figure 15 turnitin